5 Fig. 5.1 shows four rays of red light, P, Q, R and S, coming from a spotlight in a swimming pool.

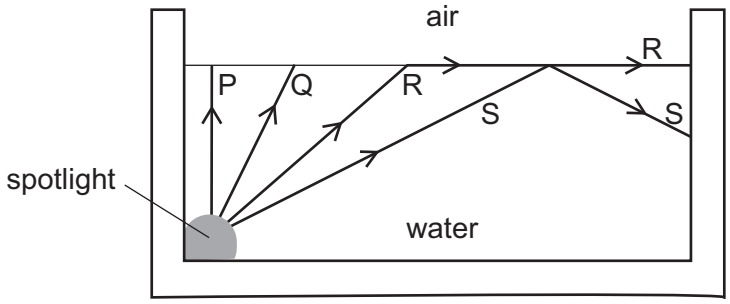

Fig. 5.1

(a) Define, in words, refractive index for a ray of light travelling from air to water.

..... [1]

(b) On Fig. 5.1, draw the path of rays P and Q at the water–air boundary. [2]

(c) The angle of incidence for ray R is 49°.

Calculate the refractive index of the water.

refractive index = ... [2]

(d) Explain why ray S is totally internally reflected at the water surface.

.....

..... [2]

[Total: 7]

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